

Elevated Levels of Cholesterol-Rich Lipid Rafts in Cancer Cells Are Correlated with Apoptosis Sensitivity Induced by Cholesterol-Depleting Agents



Lipid rafts/caveolae are membrane platforms for signaling molecules that regulate various cellular functions, including cell survival. To better understand the role of rafts in tumor progression and therapeutics, we investigated the effect of raft disruption on cell viability and compared raft levels in human cancer cell lines versus their normal counterparts. Here, we report that cholesterol depletion using methyl- β cyclodextrin caused anoikis-like apoptosis, which in A431 cells involved decreased raft levels, Bcl-xL down-regulation, caspase-3 activation, and Akt inactivation regardless of epidermal growth factor receptor activation. Cholesterol repletion replenished rafts on the cell surface and restored Akt activation and cell viability. Moreover, the breast cancer and the prostate cancer cell lines contained more lipid rafts and were more sensitive to cholesterol depletion-induced cell death than their normal counterparts. These results indicate that cancer cells contain increased levels of rafts and suggest a potential use of raft-modulating agents as anti-cancer drugs.